INFOTACT SOLUTIONS · DATA ANALYTICS INTERNSHIP

CareFlow

Clinical Pathway Process Mining

Following 1,000 patients through the emergency department, one event at a time



Agenda



Context	Problem, team, sprint plan, method
Discovery	Dataset, activities and pathway variants
Performance	Waits, rework, acuity, shifts, doctors
Modelling and BI	Feature store, dashboards
Action	Recommendations and next steps

The project at a glance

1,000	6,964	10	6
patient cases	timestamped events	clinical activities	pathway variants

23.5%	107.5 min	43.4 min	1
of patients loop back	average length of stay	average door-to-doctor	duration outlier by IQR (0.1%)

Source: main README, sagar README, day 10 audit

The problem and our objectives

The problem

- EDs struggle with boarding, bottlenecks and unpredictable turnaround.
- Redundant reviews inflate stays.
- Average length of stay hides which step is slow and where patients loop back.

Objectives

- Collect and validate a clinical event log
- Discover the common patient pathways
- Measure journey duration and waiting times
- Find bottlenecks and repeated activities
- Deliver visuals, dashboards and a prediction-ready dataset

Source: prachi README (objectives), main README

Team and contributions

AP

Akshar Patel

Lead analyst

Leakage-free imputation and scaling pipeline; 41-feature model-ready store; 16 automated checks; main README

SJ

Sagar Janathe

Analytics and BI

Notebooks for Days 3–16 (EDA to outliers); four Power BI dashboards; stakeholder summary

PB

Prachi Bhilare

Process mining

Python scripts for workflows, transitions, loopbacks and bottlenecks; process maps; recommendations

PR

Prashanth

Collaborator

Contributor on the shared CareFlow repository

Sprint timeline



Setup and clean

Environment, validation, case IDs and timestamps

Explore

Baseline EDA, demographics, activity frequency

Mine

Timelines, traces, transitions, acuity, mid-project audit

Analyse

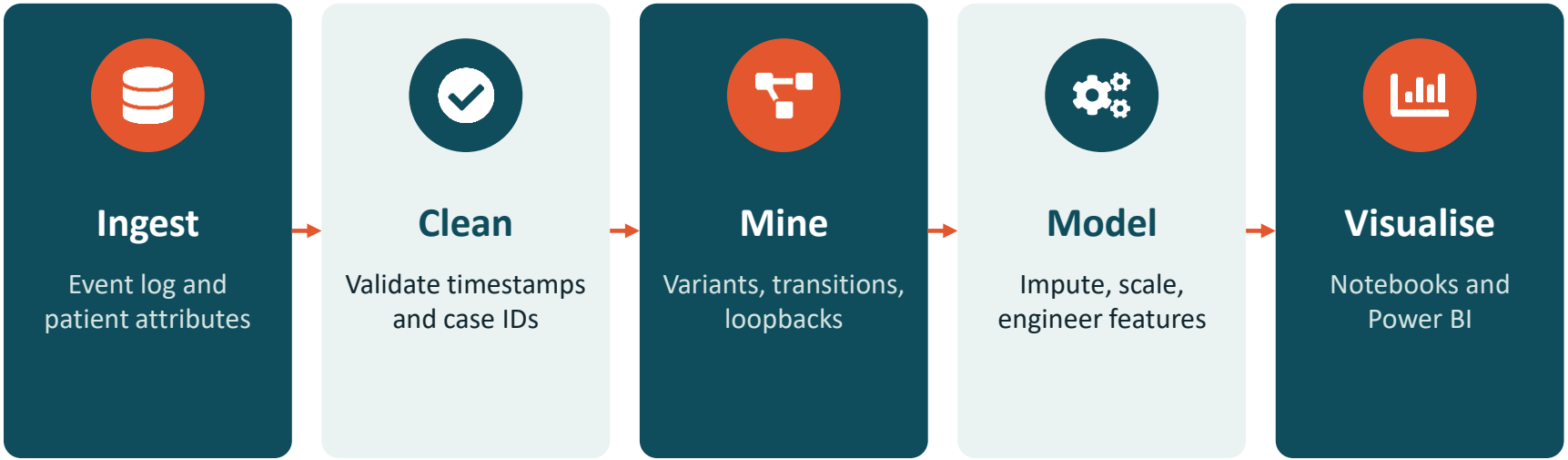
Outliers, cohorts, peak loads, feature significance, final metrics

Integrate

Code review and dashboard page integration

Source: sagar notebooks (day 3–16), day18 notes, main README sprint log

From raw log to insight



Stack: Python · Pandas · PM4Py · Scikit-learn · Matplotlib / Plotly · Jupyter · Power BI

01

Discovery

What data we have and which paths patients take





Ten clinical activities

Activity	Patients	Description
Registration	1,000	Front-desk arrival and demographics
Triage	1,000 (1,235 events)	Acuity scoring; includes 235 re-triages
Doctor Assessment	1,000	Primary physician examination
X-Ray / Radiology Result	584	Imaging and radiologist report
Lab Collection / Lab Result	371	Specimen draw and test report
Documentation Review	584	Cross-check of results and notes
Doctor Reassessment	235	Second evaluation after rework
Discharge	1,000	Formal discharge or transfer

Source: main README activity catalog

Data validation and quality

Schema

Required fields case_id, activity and timestamp checked for presence and nulls

Timestamps

Parsed to datetime, sorted per case, chronology verified

Audit

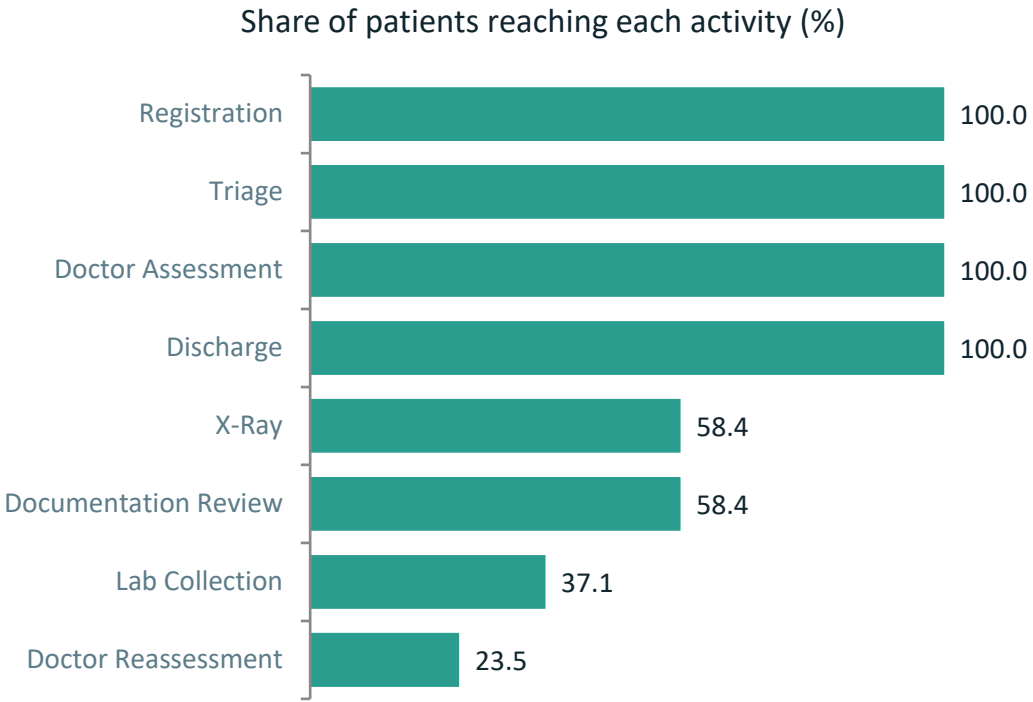
1,000 cases and 6,964 events reconciled; zero unresolved anomalies at the Day 10 audit

Pipeline

16 of 16 automated checks pass on imputation and scaling

Source: prachi docs/dataset_validation.md, sagar day 10 audit, akshar README

Who needs which diagnostics



Every journey starts at Registration and ends at Discharge.

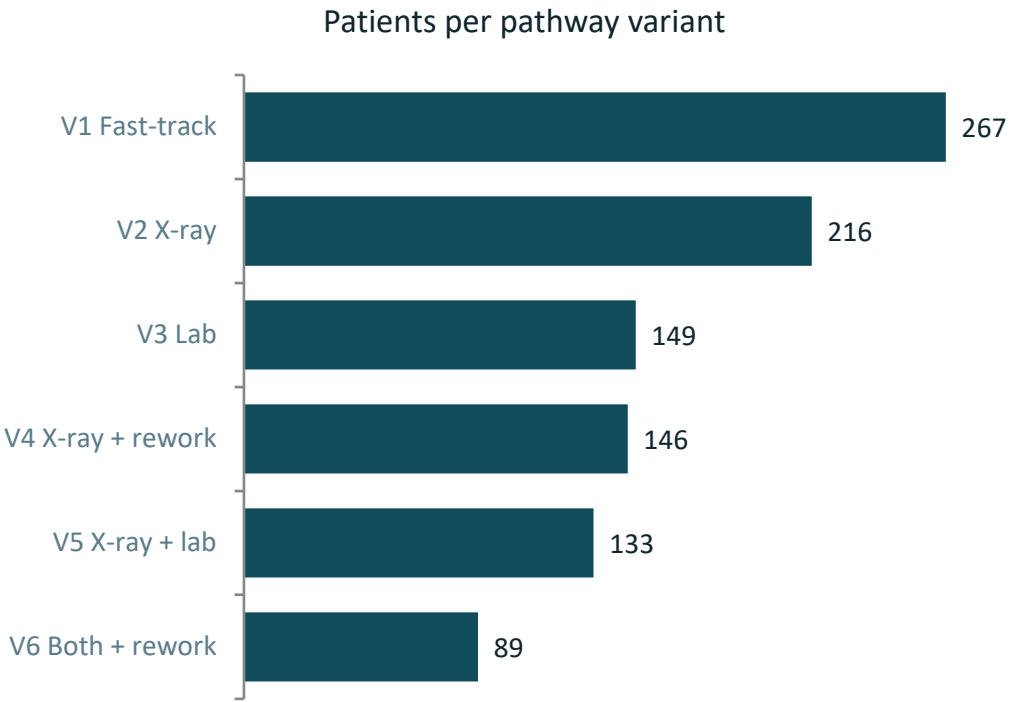
Triage fires 1,235 times: 1,000 first visits plus 235 re-triages.

26.7% go straight from doctor to discharge with no tests.

X-ray is 1.6 times as common as lab work.

Source: activity_frequency_summary.csv

Six pathways cover every patient



Average journey time

58.6 min

V1 · straight to discharge

102.3 min

V2 · X-ray only

132.8 min

V4 · X-ray with rework

174.6 min

V6 · both tests with rework

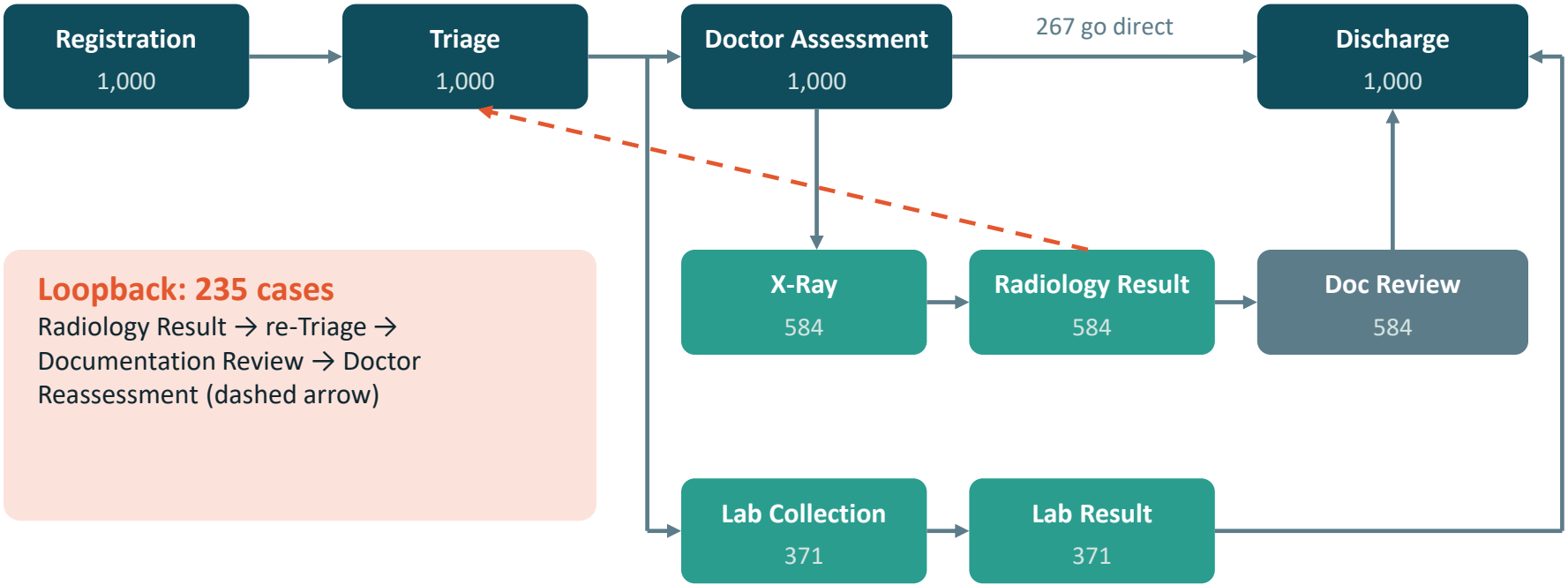
Source: pathway_variation_summary.csv

Pathway variants in detail

ID	Pathway	Cases	Share	Avg min	Range
V1	Registration → Triage → Doctor → Discharge	267	26.7%	58.6	12–107
V2	…Doctor → X-Ray → Radiology → Doc Review → Discharge	216	21.6%	102.3	52–158
V3	…Doctor → Lab Collection → Lab Result → Discharge	149	14.9%	102.9	44–157
V4	V2 plus re-Triage and Doctor Reassessment	146	14.6%	132.8	73–192
V5	X-ray track followed by lab track	133	13.3%	146.6	84–207
V6	V4 plus lab track	89	8.9%	174.6	120–234

V4 and V6 are the only variants containing a loopback; together they hold all 235 rework cases.

Patient flow map



Loopback: 235 cases

Radiology Result → re-Triage → Documentation Review → Doctor Reassessment (dashed arrow)

Source: main README (directly-follows graph), prachi process_map.svg

02

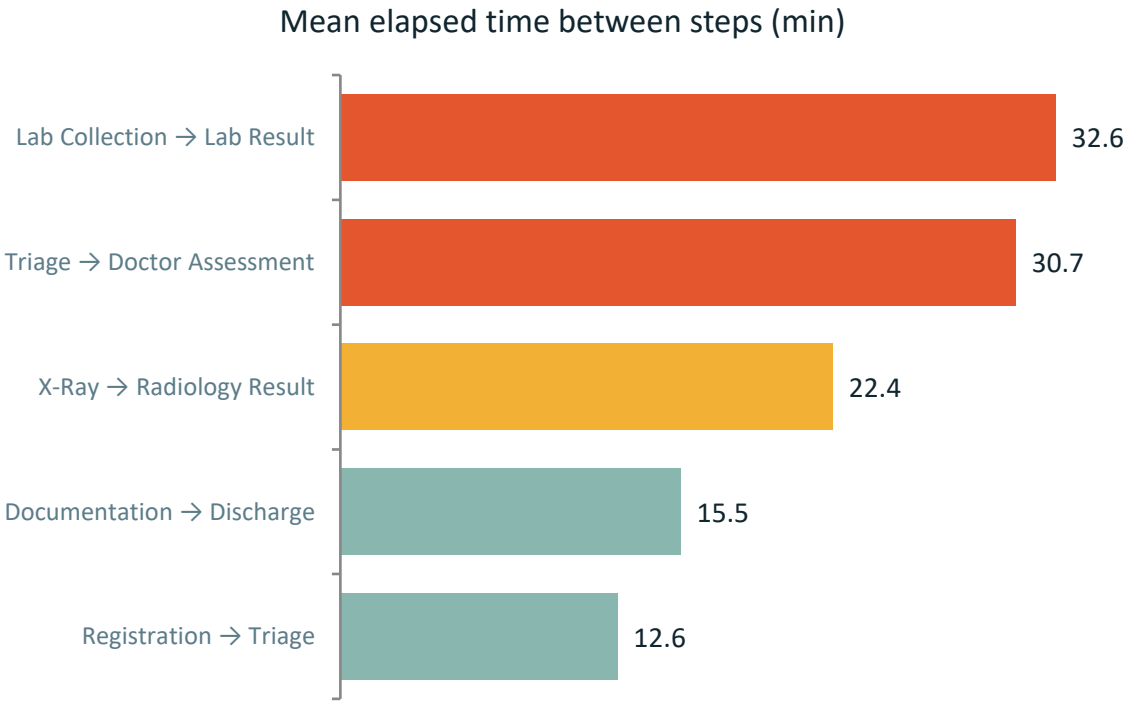
Performance

Where time is lost: waits, rework, acuity, shifts and doctors





Where patients wait



43.4 min

average door-to-doctor (12.6 + 30.7)

Lab turnaround is 45% slower than radiology, and the doctor wait is the largest delay before any diagnosis begins.

Source: workflow_performance_summary.csv

Transition times in detail

Transition	Count	Avg min	Median	Max
Lab Collection → Lab Result	371	32.6	32	50
Triage → Doctor Assessment	1,000	30.7	29	60
X-Ray → Radiology Result	584	22.4	22	35
Documentation Review → Discharge	216	15.5	16	25
Doctor Reassessment → Discharge	146	15.4	16	25
Doctor Assessment → X-Ray	584	15.1	15	25
Lab Result → Discharge	371	14.9	14	25
Registration → Triage	1,000	12.6	12	25

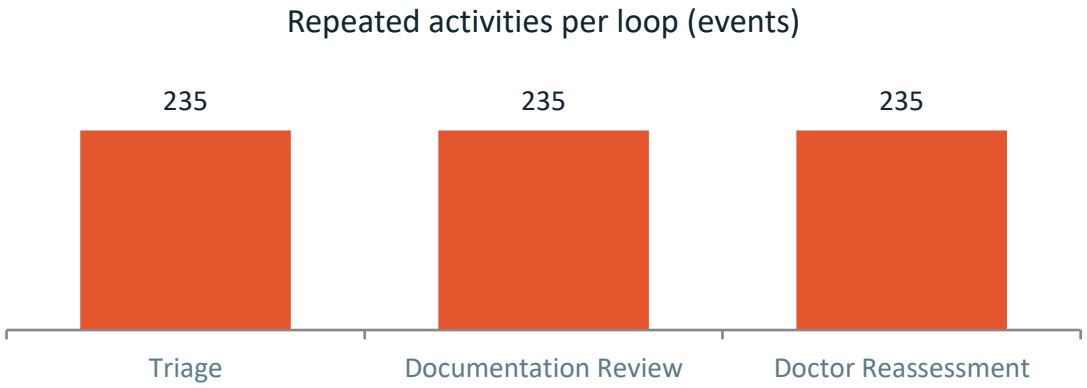
Source: workflow_performance_summary.csv (5,964 transitions, 16 types)

Rework: nearly 1 in 4 loop back

23.5%
of cases loop back

235
cases

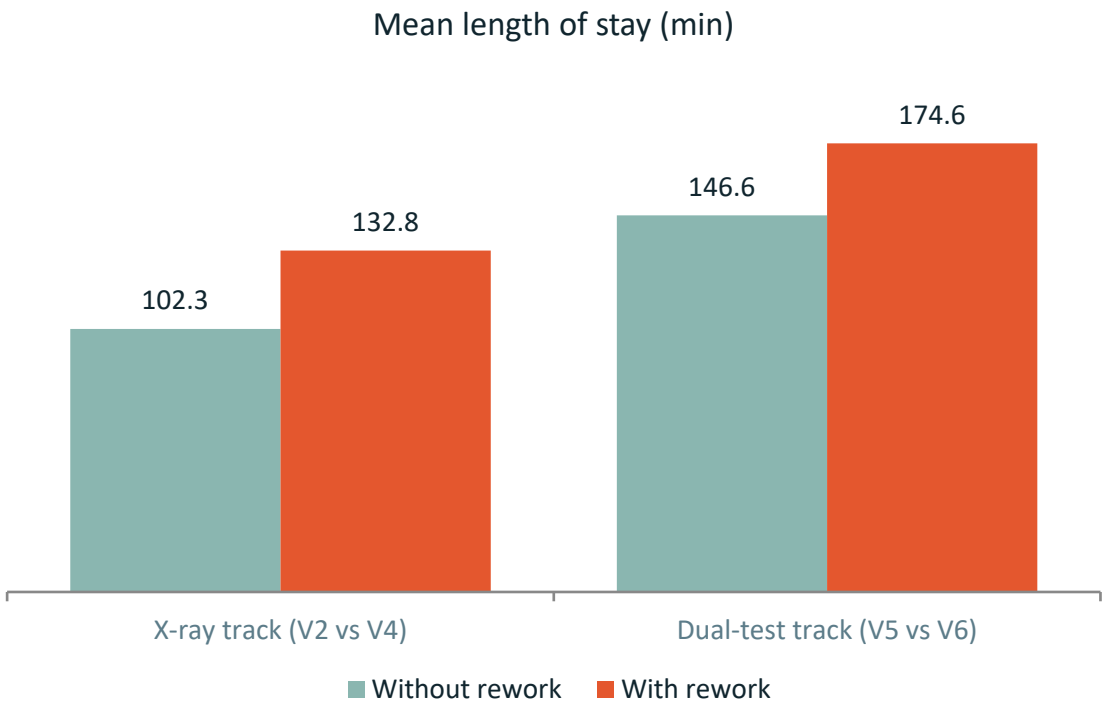
705
loopback events



Each loop is a repeat Triage, a Documentation Review and a Doctor Reassessment, always after a radiology result. Only variants V4 and V6 contain loops, and the median loopback wait is 35 minutes in the ED bay.

Source: loopback_activity_summary.csv, akshar README

Rework adds 28 to 30 minutes



+30.5 min

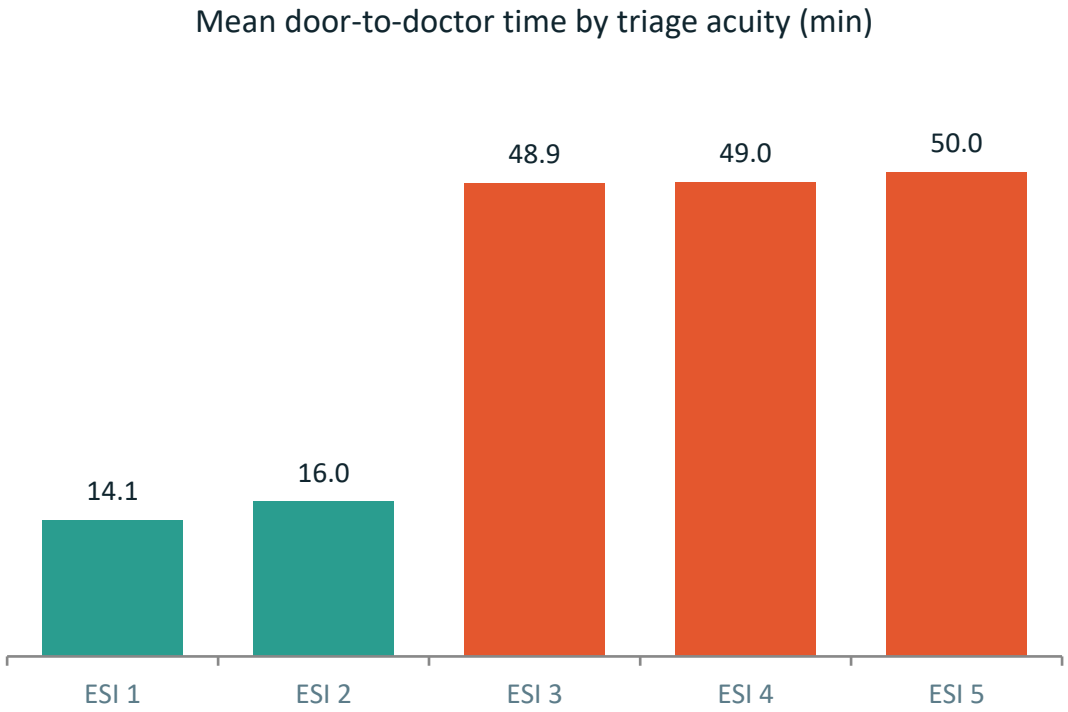
+29.8% on the X-ray track

+28.0 min

+19.1% on the dual-test track

Source: akshar README (rework penalty analysis)

Acuity splits care in two



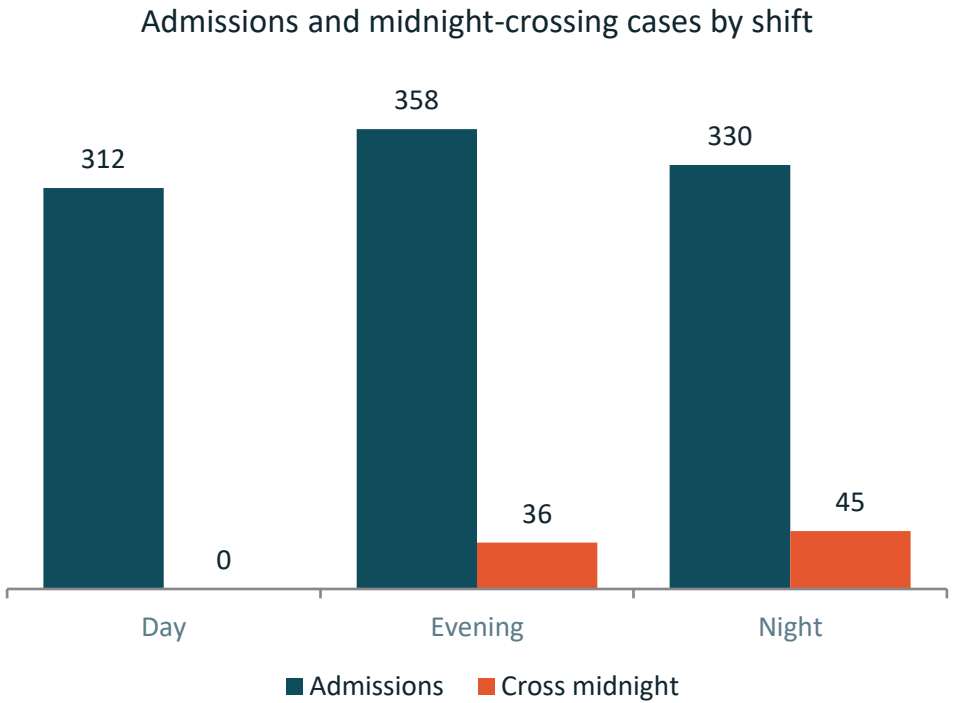
ESI 1–2 (17.0% of patients)
Seen in about 16 minutes; triage in under 5.

ESI 3–5 (83.0% of patients)
Seen in about 49 minutes, roughly 3 times longer, as stable patients queue.

Acuity is the top delay driver in the feature analysis.

Source: akshar README (ESI triage stratification)

Shift dynamics and midnight boarding

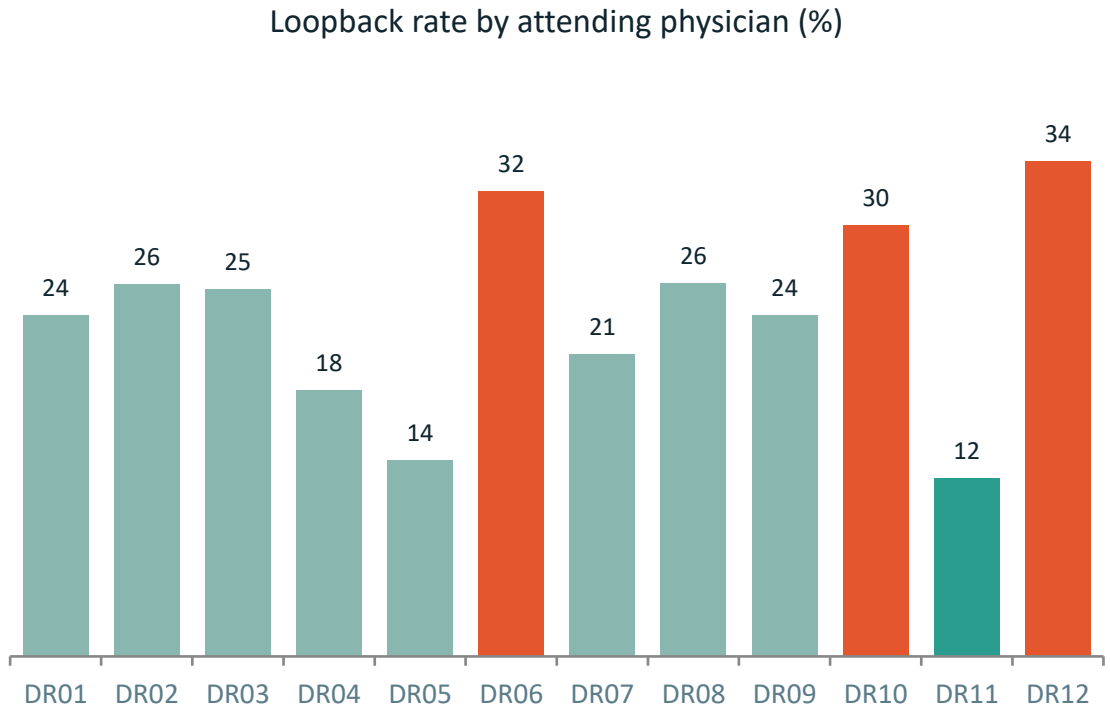


Shift	Mean LOS	Loopback
Day 07–15	110.6	25.6%
Evening 15–23	105.1	21.5%
Night 23–07	107.1	23.6%

All 81 midnight-crossing patients arrived in the Evening or Night shift; none arrived on Day shift.

Source: akshar README (shift dynamics)

Physician variation



12% to 34%

DR11 reworks 12.3% of cases, DR12 34.1%.

Mean stay differs too: DR01 101 min, DR12 116 min.

Doctor assignment is a useful predictor for rework.

Source: akshar README (physician workload analysis)

03

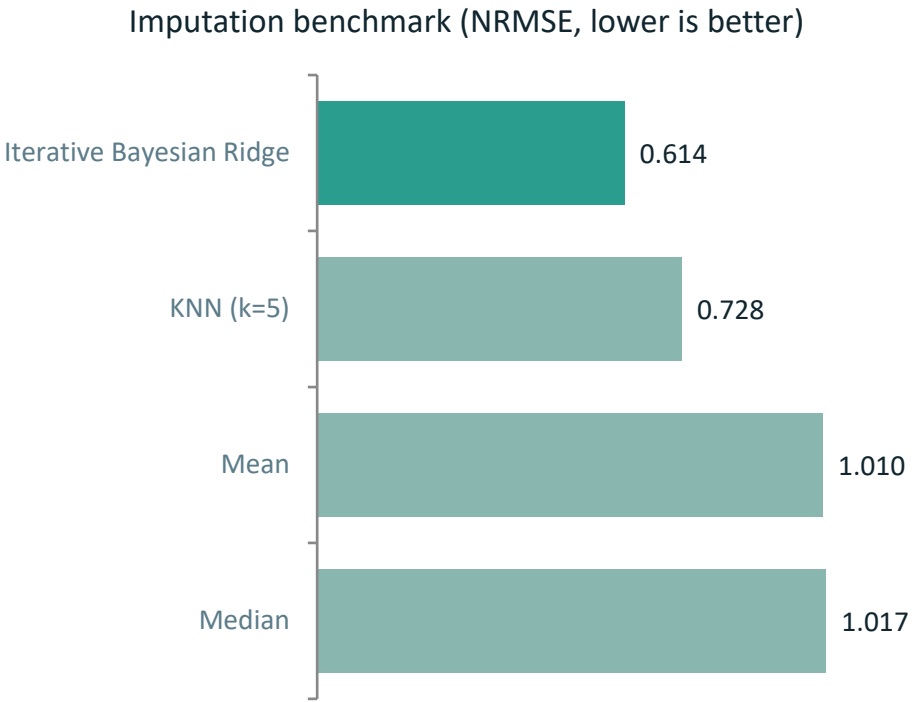
Modelling and BI

Leakage-free data for prediction, dashboards for managers





A leakage-free pipeline



Chronological split
800 train / 200 test cases, cutoff 18 Feb 2026 11:00

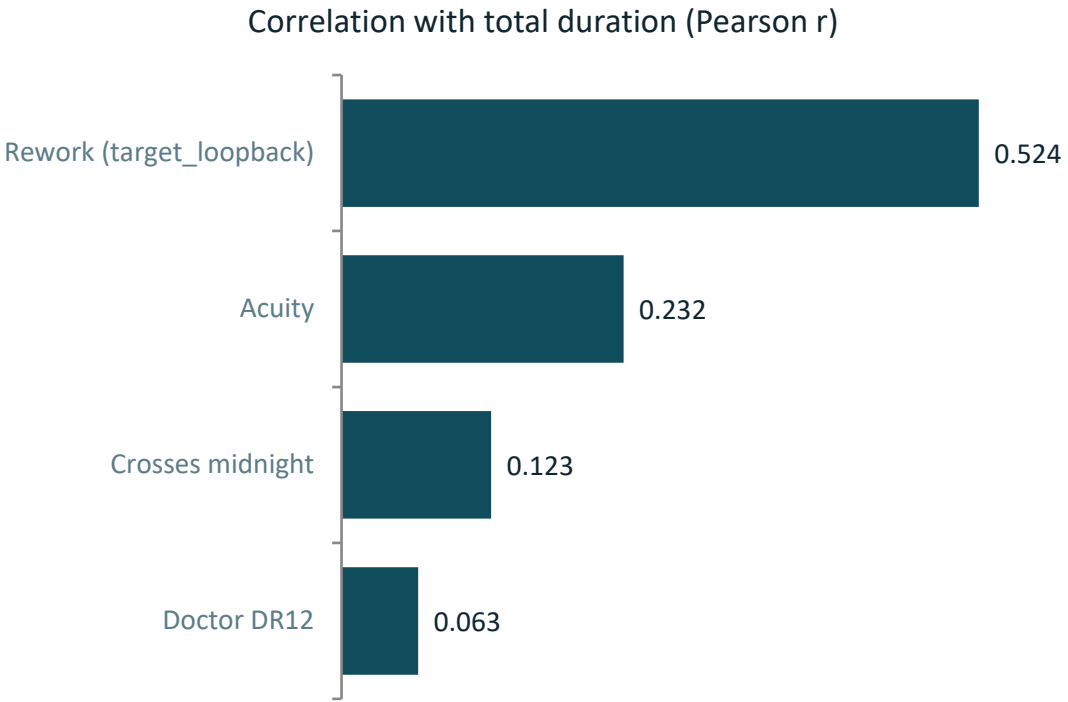
Structural gaps flagged
has_xray, has_lab, has_reassessment; no invented procedures

Dual scaling
Z-score and MinMax fitted on train only; parameters saved to JSON

Feature store
41 arrival-time features incl. sin/cos hour; 16 of 16 checks pass

Source: akshar README, imputation_benchmark.csv

What predicts length of stay



Rework is the strongest driver (r = 0.52), then acuity (0.23). Age and doctor DR01 are near zero.

Arrival hour harmonics best predict midnight crossing (cos r = 0.39).

Both targets are knowable at admission, so models can flag risk early.

Source: akshar README (target correlation analysis)

Power BI dashboards

PAGE 1

Emergency Overview

LOS 107.5 min across 1,000 encounters

PAGE 2

Process and Bottlenecks

Transition delays and flow

PAGE 3

Doctors and Workloads

Physician volume and rework

PAGE 4

Loopbacks and Outliers

Rework paths and duration outliers

The Python branch adds an interactive workflow dashboard and SVG process maps.

Source: sagar/dashboard, prachi/docs

Recommendations

1	**Cut door-to-doctor wait**	Dynamic triage routing and fast-track rooms for non-complex cases
2	**Speed up lab turnaround**	Point-of-care testing for high-acuity patients
3	**Reduce rework (23.5%)**	Standard diagnostic checklist at first doctor assessment
4	**Staff for the evening surge**	Arrival-hour forecasting to cover handovers and midnight boarding
5	**Predict length of stay**	Feed the scaled feature store into bed-capacity forecasts

Key takeaways

- Waiting drives the stay: doctor and lab waits top the delay list.
- Nearly 1 in 4 patients loop back, adding about 30 minutes.
- Acuity creates a 3x wait gap between critical and stable patients.
- The leakage-free feature store is ready for stay and rework prediction.

Next: train and validate stay and rework models, then deploy live process-mining dashboards.



Thank you

Questions and discussion

Akshar Patel · Sagar Janathe · Prachi Bhilare · Prashanth

